16 Using $x_{n+1} = -2 - \dfrac{4}{x_n^2}$

with $x_0 = -2.5$

(a) find the values of x_1, x_2 and x_3

Use the previous unrounded value each time.

$$x_1 = -2 - \frac{4}{(-2.5)^2} = -2.64$$

1 mark
Substitute $x_0 = -2.5$

$$x_2 = -2 - \frac{4}{(-2.64)^2} = -2.573921028...$$

$$x_3 = -2 - \frac{4}{(-2.573921028...)^2} = -2.603767255...$$

1 mark
Substitute successively to find x_2 and x_3

1 mark
All three values correct

$x_1 =$ -2.64

$x_2 =$ -2.57392

$x_3 =$ -2.60377

(3)

(b) Explain the relationship between the values of x_1, x_2 and x_3 and the equation $x^3 + 2x^2 + 4 = 0$

Rearrange $x^3 + 2x^2 + 4 = 0$:

$x^3 = -2x^2 - 4$, so $x = -2 - 4/x^2$.

1 mark
This is the equation in iterative form

The iterates approximate a solution of the equation.

They converge towards about -2.59.

1 mark
Successive estimates approach a root

(2)

(Total for Question 16 is 5 marks)